



Become the next unicorn with artificial intelligence

How 8 startups are using AI to drive growth and gain competitive advantage



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Startups in the age of accessible artificial intelligence

Thanks to the rapid adoption of cloud computing, the rise of compute power and data volumes, and the emergence of easy-to-use solutions that require little or no experience, artificial intelligence (AI) is now more accessible than ever.

Startups at any stage are leveraging AI to achieve transformative business results. For many, AI has become a core component of their operations. According to IDC, spending on AI in the United States will grow to \$120 billion by 2025, representing a compound annual growth rate (CAGR) of 26 percent over the 2021–2025 forecast period.¹

Amazon Web Services (AWS) is playing a pivotal role in the advancement of AI, empowering startup customers to effectively use and derive the most benefit from the technology. These startups span fintech, healthtech, ecommerce, gametech, and technology.

Categorizing artificial intelligence objectives

By offering the broadest and most complete set of AI services, AWS is able to meet startups wherever they are in their ML journey and help them achieve their business objectives. More than 100,000 customers use **AWS AI services** for everything, from enhancing the customer experience and predicting business outcomes to reducing fraud and start with accelerating innovation.

AWS helps startups across industries successfully leverage AI across three categories of critical business objectives: accelerating innovation, improving the customer experience, and optimizing business operations.

This eBook features success stories from AWS startup customers that have achieved transformative results across these three categories using powerful AI capabilities.



SUCCESS STORIES

Accelerating innovation

Startups also leverage AI to help accelerate innovation across their products, services, and applications. This can enable your team to accomplish more— regardless of their machine learning (ML) skill level—while driving efficient resource management and helping to keep costs low.

Read on to discover how Frollo, Synchronized, and Guru Technologies are leveraging AI to achieve transformative results for their business.





Opening the door to open banking

Frollo is Australia's first open banking intermediary. This purpose-driven fintech is on a quest to help people feel good about money. To date, Frollo has made over 25 million open banking API calls.

Opportunity

Frollo wanted to enable banks to go beyond their current capacity and develop a real engagement model that provides more money management and budgeting features for customers. The Frollo Data Enrichment API provides data enrichment solutions such as categorizing customer transactions and identifying merchants. This API and supporting ML models sort consumer spending into more than 60 categories, such as travel, taxes, and groceries.

Solution

Cloud-based ML has been instrumental in accelerating the development of the Frollo platform, and **Amazon SageMaker** has been a core element of its ML strategy. Frollo is now running most workloads on AWS, having chosen the platform for its data and analytics capability, built-in security controls, and ML tools. With SageMaker, Frollo's team of data scientists can build their own ML algorithms and retain control over training their datasets.

Results

- **Improved accuracy:** Frollo achieved a 95% accuracy rate for ML models that categorize transactions
- **Better performance for customers:** Frollo enables banks to return credit results in only 20 seconds instead of 3–5 minutes
- **Improved security and reduced risk:** Frollo solutions secure product reference data for banks and recuse risk in loan origination

“With Amazon SageMaker, we’ve never had an issue with application limits or performance. We’re now measuring an end-to-end latency for transaction categorization and merchant identification of 1,500 milliseconds.”

Tony Thrassis, CIO, Frollo



Automating content creation and segmentation on the AI-enabled video platform

Synchronized transforms linear and passive videos into smart videos with the help of AI that understands video content and enriches it with metadata. The Synchronized platform automates entire processes for the transformation of linear TV content into digital programs that enhance viewer experiences.

Opportunity

The company rolled out its video automation solution on MYTF1, the on-demand video platform of TF1, France's leading free-to-air TV channel. Before working with Synchronized, TF1 was unable to automatically generate thumbnails for each program on MYTF1 or guarantee the quality of the thumbnails based on specific editorial criteria, requiring the editorial team at TF1 to manually process hundreds of hours of content every month. The Synchronized platform's Smart-Thumbnails service automates all the tasks involved in creating and selecting video thumbnails.

Solution

Powered by the Synchronized platform, all video images are automatically analyzed and processed using **Amazon Rekognition** to generate a significant mass of data, such as the recognition of faces, emotions, objects, and text. This mass of data is subjected to the editorial guidelines predefined in the platform. Images corresponding to the guidelines are selected as potential images, with the final recommendation determined by the algorithm. TF1 editors can access all suggestions in Synchronized Studio in three clicks and choose alternate images.

Results

- **Elevates content quality:** Synchronized users can guarantee the quality of thumbnails based on specific editorial criteria
- **Automates video thumbnails:** The solution helps automate and accelerate the creation of more than 500 hours of programs every month
- **Liberates staff time:** Saves hundreds of hours of editors' time every month

"The beauty of it all is that a startup can embark on highly ambitious projects while building on and collaborating with companies like AWS, which provide access to the results of their work and research. That was impossible 15 years ago. Today, these synergies are unleashing innovation."

Guillaume Doret, CEO, Synchronized



Creating new business opportunities with improved search results

Guru Technologies provides knowledge management software to make it simple for companies to access their internal information whenever needed, no matter where it's stored.

Opportunity

Experiencing growth in providing fast, relevant query results, Guru was looking to increase scalability, speed, and reliability to continue innovating at scale. It was seeking a managed service to make it easy to perform interactive log analytics, real-time application monitoring, website search, and more.

Solution

Guru initiated its migration to **Amazon OpenSearch Service** in 2020. Within a short amount of time, the startup saw several benefits from the migration. For one, Guru was able to use **Amazon EMR**—an industry-leading cloud big data service for processing vast amounts of data using open-source tools—to develop an experimentation framework for improving the search result relevance of its search engine. This would ultimately help users find the information they're looking for more quickly.

By migrating from self-managed Elasticsearch clusters to Amazon OpenSearch Service, Guru was able to spend more time focusing on experimentation and innovation. With this new framework, Guru has a scalable path for experimenting with machine learning and deep learning, including implementing the k-nearest neighbors algorithm and learning to rank.

As a startup looking to grow quickly while releasing new features, Guru found that AWS could provide the reliability, scalability, and elasticity the company needed to continue to innovate.

Results

- Improves the search service for users more quickly
- Reduces time and resources spent on Elasticsearch
- Runs experiments that involved replaying nearly half a billion queries
- Reduces experimentation time from weeks to hours
- Improves search relevancy by 10%
- Allows for more time to focus on innovation

“One thing that’s great about AWS is that it’s self-service: you can move as fast as you want within the environment itself....You can get in really quickly, use whatever systems are necessary for your startup, and then just iterate and iterate and iterate. You can build everything up without needing permission to engage with a third-party vendor that might slow you down.”

Steve Mayernick, Director of Product Marketing, Guru Technologies

Machine learning services for accelerating innovation

Amazon SageMaker offers an end-to-end service to build, train, and deploy ML models for any use case with fully managed infrastructure, tools, and workflows



Make machine learning more accessible

Enable more people to innovate with ML through a choice of tools—integrated development environments (IDEs) for data scientists and no-code visual interfaces for business analysts.



Prepare data at scale

Access, label, and process large amounts of structured (tabular) data and unstructured data (photos, videos, and audio) for ML.



Accelerate machine learning development

Reduce training time from hours to minutes with optimized infrastructure. Boost team productivity up to 10 times with purpose-built tools.



Streamline the machine learning lifecycle

Automate and standardize MLOps practices across your business to build, train, deploy, and manage models at scale.

SUCCESS STORIES

Improving the customer experience

Many startups are turning to the intelligent, predictive powers of AI to enhance the customer experience and improve customer engagement. If customer service and support are critical for your business, you're likely to find practical inspiration in the following three success stories. Read on to discover how Paytm, Showpad, and Segment are demonstrating the transformative power of AI to elevate customer satisfaction scores.





Boosting homepage sales with personalized recommendations

Paytm is a financial services company for the consumer brand of India's mobile internet company, One97 Communications. It offers full-stack payment and financial solutions to millions of consumers and merchants.

Opportunity

Paytm was looking to leverage personalized recommendations on the homepage of its site to increase the sales and click-through rates of its ecommerce service, Paytm Mall. The company sought a solution that could analyze and propose recommendations to over 10 million daily active users who shop for retail goods on Paytm Mall.

Solution

To generate recommendations for each customer, Paytm turned to AWS for a fully managed, robust ML solution, Amazon Personalize. Leveraging the same ML technology used by Amazon.com for near real-time personalized recommendations, the service would be quicker to implement with a better turnaround time than creating an in-house solution. Paytm leverages Amazon EMR, a cloud big data service for running large-scale distributed data processing jobs, to process user data.

Results

- Delivers a 5.5%–6% conversion rate from the Paytm Mall homepage
- Increases total orders and volume of sales from the homepage
- Produces a higher conversion rate than with any other recommendations model
- Implements beta ML models in half the time of building in-house solutions

“Personalization helps users get what they need in the fewest clicks possible. By having personalization using Amazon Personalize, we create a unique app for every user based on individual choices and preferences.”

Ankur Gogate, Technical Lead, Paytm



Helping sellers become trusted advisors to buyers

Showpad is a global leader in sales- and revenue-enablement technology. It integrates industry-leading training and coaching software with innovative content solutions to empower sales and marketing teams to engage the modern buyer.

Opportunity

Generative AI allows buyers to access information that only salespeople previously had access to in order to inform their purchasing decisions. To meet buyer expectations, Showpad needed to innovate their seller roles to help prospects and customers make sense of information in the context of the customers' specific business challenges.

Solution

Showpad embedded AI innovation throughout the product experience and was able to rapidly experiment and subsequently push new models to production by relying on **Amazon Bedrock** and **Amazon SageMaker**. PitchAI, Showpad's latest sales coaching software, helped improve pitch quality with actionable feedback in real time. PitchAI's AI-powered search enriched search results, reducing sellers' time spent looking for the information. Its AI-powered test questions also streamlined the process of creating training materials for sellers.

Results

- **Improved experience:** Showpad was able to add value during the sales process and build buyer trust and confidence
- **Automated workflow:** Showpad relieved time-consuming administrative tasks, allowing salespeople to focus on innovation
- **Enhanced decision making:** Showpad was able to drive better conversations that empowered more effective decision making in solving customer challenges

"Showpad is leveraging AI to improve how sales and marketing teams partner together in pursuit of a shared goal—closing more deals. It enables sellers to focus more time on what truly matters—bringing meaningful value to each buyer interaction."

Tony Grout, CPO, Showpad



Creating custom recommendations to empower customer experiences

Segment is a customer data infrastructure company that helps its clients collect and unify their data. Using this data and AWS AI solutions, Segment allows organizations to deliver personalized recommendations that create meaningful customer experiences.

Opportunity

Segment recognized that many of its customers did not have the data necessary to power and perform personalized recommendations. To meet this need, the company set out to provide a central hub in which to collect, train, and track all data, as well as run an ML pipeline. With this system, Segment helps its clients deliver product and content recommendations based on their customer data.

Solution

To help run workloads and process over 450 billion events per month, translating to roughly 250,000 requests per second, Segment turned to the power of AWS. Its solution involved using thousands of **Amazon Elastic Compute Cloud** (Amazon EC2) instances and running more than 16,000 Docker containers on **Amazon Elastic Container Service** (Amazon ECS). This data feeds into **Amazon Personalize**, resulting in improved customer experiences.

Results

- **More comprehensive insights:** Segment generated insights from all data across all apps
- **Improved customer experiences:** Segment discovered its customers' needs and translated the data easily into recommendations and world-class experiences
- **Faster growth:** Segment was able to move fast to propel business growth

"It's a beautiful synergy where they can spin up Segment easily from day one and get going collecting all their data. Then, they can use that data to power recommendations without having to build out their own machine learning pipeline using Amazon Personalize."

Calvin French-Owen, CTO & Co-Founder, Segment

IMPROVING CUSTOMER EXPERIENCE

Solutions to improve customer experience

Virtual agents and chatbots

[Amazon Bedrock](#)

[Amazon Polly](#)

[Amazon Lex](#)

AI-enabled contact center

[Amazon Connect](#)

[Amazon Transcribe](#)

[AWS Contact Center
Intelligence \(CCI\)](#)

[Call Analytics](#)

Personalization

[Amazon Bedrock](#)

[Amazon Personalize](#)

Amazon SageMaker—Prepare data, and build, train, and deploy machine learning models

Develop your own models with the end-to-end service SageMaker

Choose from pre-built models and purpose-built tools for every stage of the AI lifecycle

Access no-code tools for business analysts to generate ML predictions

SUCCESS STORIES

Optimizing business operations

With its ability to process massive amounts of data quickly and intelligently, AI can help startups streamline operations, automate processes, and predict business outcomes. This can help improve productivity and augment team capabilities. The following success stories from TWAICE and PureTech Global demonstrate how core business operations can be optimized with the power of AI.



Extending the life of batteries with unified infrastructure and predictive analytics

TWAICE combines specialized battery knowledge and AI to provide detailed and actionable battery insights for electric vehicle (EV) makers and renewable energy providers. Its predictive analytics platform eliminates risk around health, safety, and warranty issues, empowering TWAICE customers to outperform their peers.

Opportunity

This software startup is unifying predictive analytics with Internet of Things (IoT) data to help EV and renewable energy businesses optimize their use of battery technology across their vehicles or energy storage systems. Of course, performing serious analytics requires extensive compute power to process large amounts of data, which has increased from 10,000 items a week to several billion a day. TWAICE's on-premises servers and databases would have made this task slow and expensive, hindering its growth as well as its ability to provide actionable results.

Solution

TWAICE increased the speed and accuracy of its analytics by 300 percent with its move to the AWS Cloud-native platform. In addition to increasing IT team efficiency, the successful migration helped reduce infrastructure development time from several months to two weeks and cut deployment times from three days to a few hours. That has allowed the total number of TWAICE customers to nearly double and the amount of data processed to increase 100,000-fold. With its unified IT infrastructure running on AWS, the company can focus on its core business: enabling better battery technology.

Results

- **Quick cloud migration:** TWAICE completed the migration from its on-premises infrastructure to a cloud-based platform within 6 months (without having to increase the size of its DevOps team)
- **Fast, functional onboarding:** Reduced overall onboarding time for new customers from up to 2 weeks to only a few hours
- **Rapid, accurate results:** The platform can now deliver analytics results in minutes—results that are up to 300% more accurate than before

“By building on AWS, we’ve reduced our IT complexity. Now, we can focus on what we’re best at—helping our customers get the most out of their battery technology.”

Jeff Glasson, CTO, TWAICE



Artificial intelligence optimizes billing revenue

PureTech Global is a mobile advertising and content provider. Based in Malaysia, it offers telecommunications subscribers in 13 countries a range of value-added, over-the-top (OTT) content such as music, games, and wallpaper.

Opportunity

To take advantage of AI and ML, PureTech turned to AWS solutions. They were looking to leverage consumer data and deliver more personalized, impactful apps to increase revenue for the company and its telco clients.

Solution

PureTech engaged with an AWS Premier Consulting Partner and AWS solutions to leverage AI-generated insights from past subscription data. **Amazon Forecast** helps predict the optimal times to send renewal notices to its customers for each of PureTech's apps. PureTech has a data lake built on **Amazon Simple Storage Service** (Amazon S3) to store data and **AWS Lambda** provision compute resources for analysis and forecasting. **AWS Glue** integrates the data, and **AWS AppSync** develops APIs to automatically send renewal notices to customers.

Results

- Increases of 20% in its billing renewal success rate
- Enables a granular level of data insights and reporting
- Runs AI-enabled billing system with low overhead
- Evolves AI model over time according to new data inputs
- Improves sales and marketing reports to monitor business performance
- Expands its international presence by extending its AI billing system to other countries

“Even in the first few months post-implementation, the AI project has generated more revenue for us. This, in turn, allows us to provide more value-added services and quality content to subscribers.”

John Lim, CTO, PureTech Global

Solutions to optimize business operations

Intelligent document processing

[Amazon Augmented AI](#) (Amazon A2I)

[Amazon Comprehend](#)

[Amazon Comprehend Medical](#)

[Amazon Rekognition Content Moderation](#)

[Amazon SageMaker](#)

[Amazon Textract](#)

Fraud detection

[Amazon Fraud Detector](#)

[Amazon SageMaker](#)

Solutions to boost employee productivity

Content discovery

[AWS Enterprise Q](#)

Content creation

[Amazon Bedrock](#)

[AWS Enterprise Q](#)

Text summarization

[Amazon Bedrock](#)

[AWS Enterprise Q](#)

Prepare data and build, train, and deploy machine learning models with Amazon SageMaker

Startups that want to develop their own models can execute against any of the above use cases with [SageMaker](#). The end-to-end service provides pre-built models and purpose-built tools for every stage of the AI lifecycle—and even includes no-code tools to generate ML predictions.

NEXT STEPS

Start realizing the business value of artificial intelligence today

A growing number of startups have leveraged the most comprehensive selection of AI services from AWS to solve their business challenges and accelerate their AI journeys. Choose the AI services that help your startup go beyond seed to growth and expansion—and learn how to move forward today.

Accelerate your AI journey with AWS ›

Vist the AWS Startups generative AI hub ›

